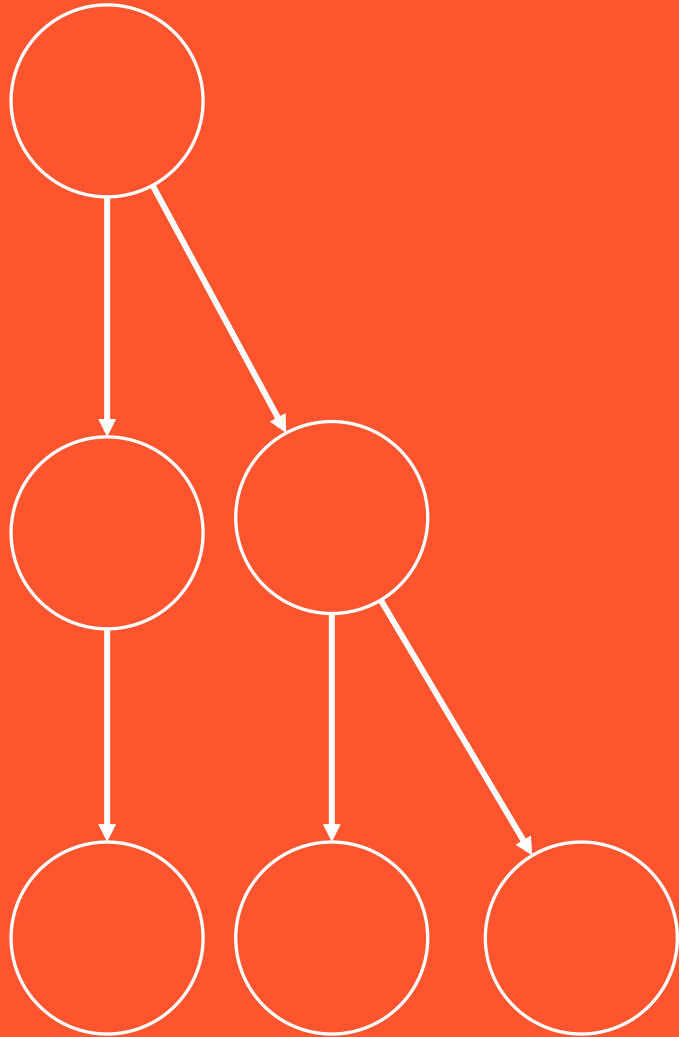




BCOG 100

Vertebrate Neural Architecture
Lecture 6.2 — The Division of Labor



UNIVERSITY OF
ILLINOIS
URBANA - CHAMPAIGN

To Start: What Would Count as Evidence?

In your group, three minutes.

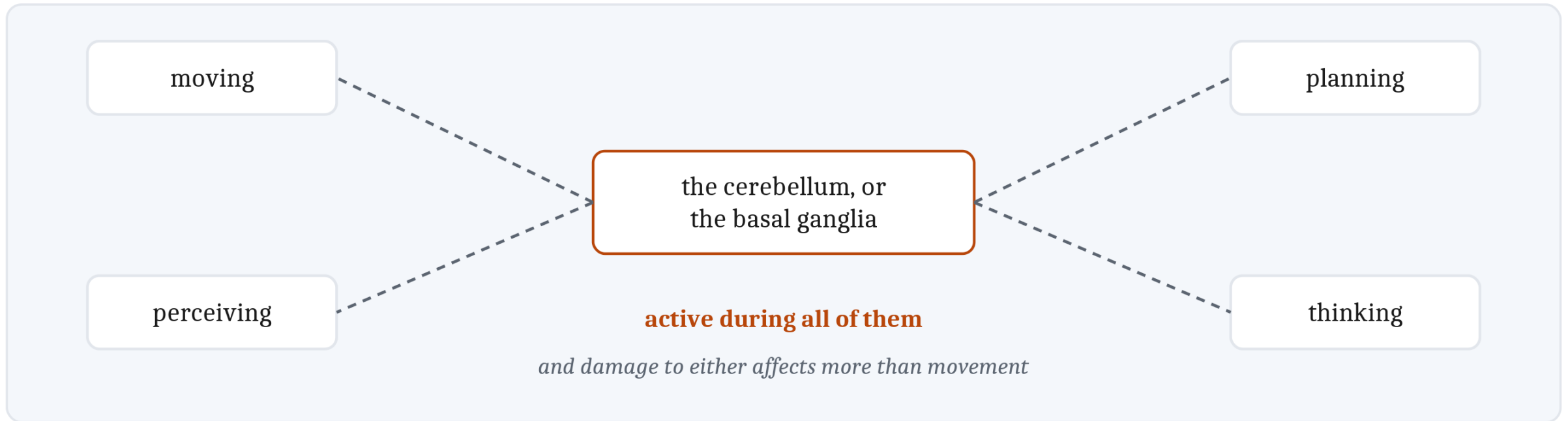
The last lecture ended on a claim: whether a brain gives two jobs two separate structures is an empirical question.

Suppose we wanted to know whether one part of the brain is *for* moving. What evidence would be convincing?

What result would cast doubt on it?

What Is Each Part For? The Wrong Question

Active during movement, perception, planning and thought: going from that activity to a job is **reverse inference**.



Reverse inference: the region was active during the task, so the task used that region's process.

Module 2
a region, in brain imaging

Module 5
a mechanism: learning and development

Module 6
a structure, in recordings

A region active during many different things licenses very little on its own.

A Better Question: How Does Each Part Learn?

Doya's proposal: each structure is specialized in a method, not a job. And a test for which level a claim is at.

Algorithmic

would hold of a system built
out of something else entirely

**hold every action back,
and release one**

a vertebrate

a robot

a simulation

Implementational

turns on which cells, what chemicals,
and where they sit

**the basal ganglia do it
with dopamine and
two opposed pathways**

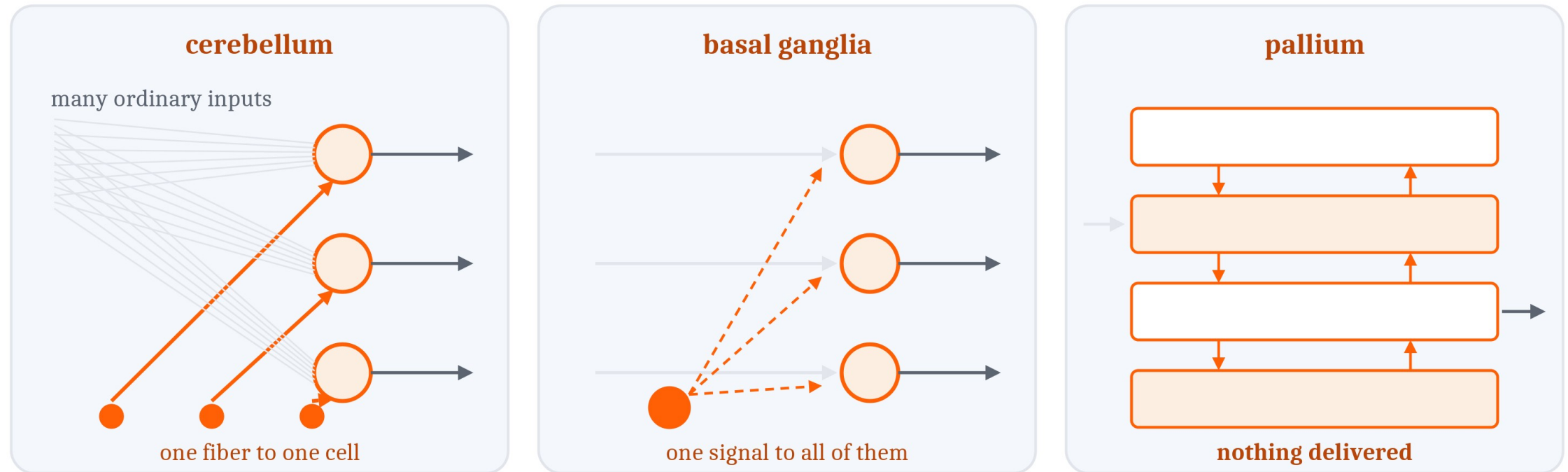
*true of one kind of animal's brain,
and of nothing built differently*

Most of this lecture: algorithmic claims, stated over parts named implementationally.

Two Levels of a Single Change

Each family of learning pairs an animal-level problem with what has to reach a synapse. Anatomy can only be asked about the second half.

Three structures, three kinds of delivery

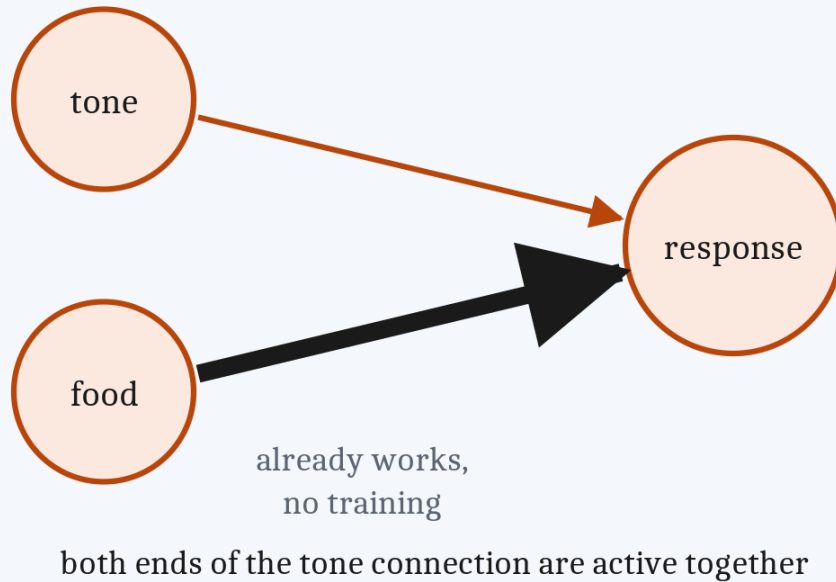


the arrows below each structure are what has to be delivered to it, not what it computes

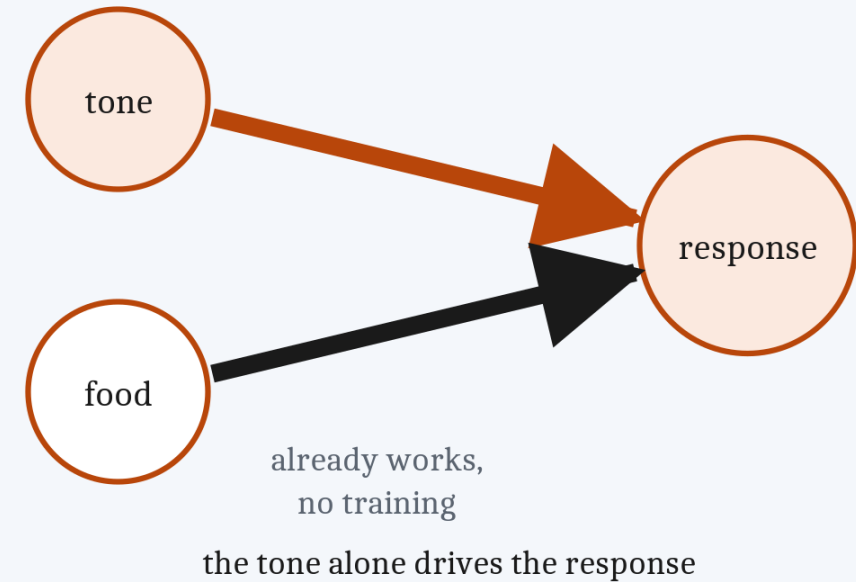
Learning Without a Teaching Signal

Unsupervised learning needs only the two cells a connection joins. The pallium gets it not because it happens only there, but because nothing else arrives there.

While tone and food are paired



After enough pairings



Nothing in the rat had to know that the tone predicted food, or that anything was being learned.

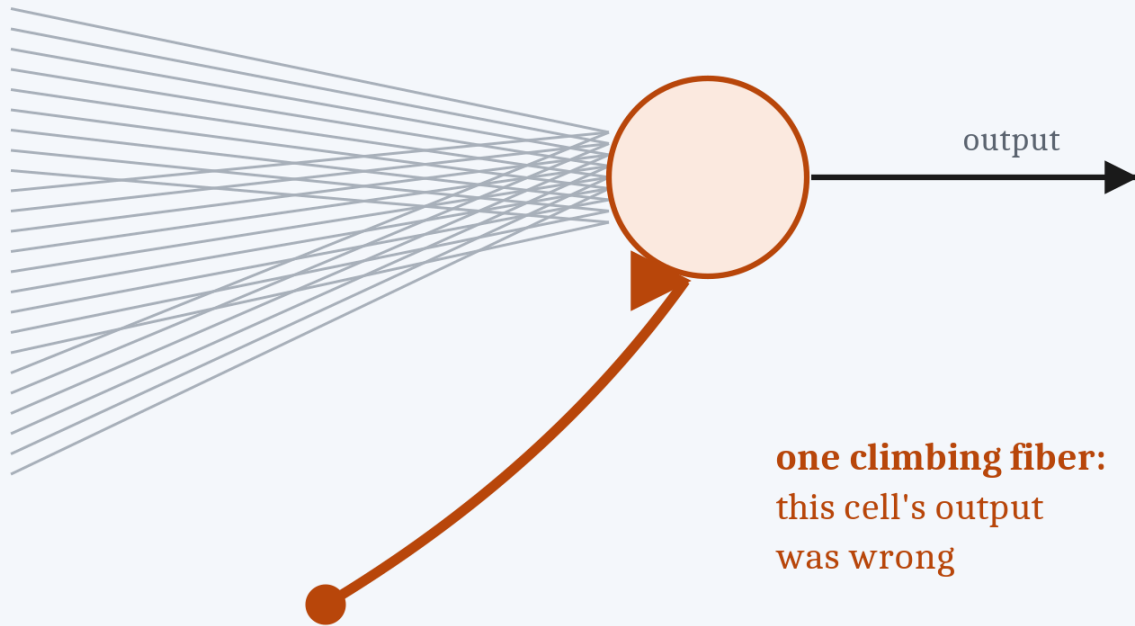
Nothing has to be delivered, so no structure is needed to deliver it.

Learning from an Error, Output by Output

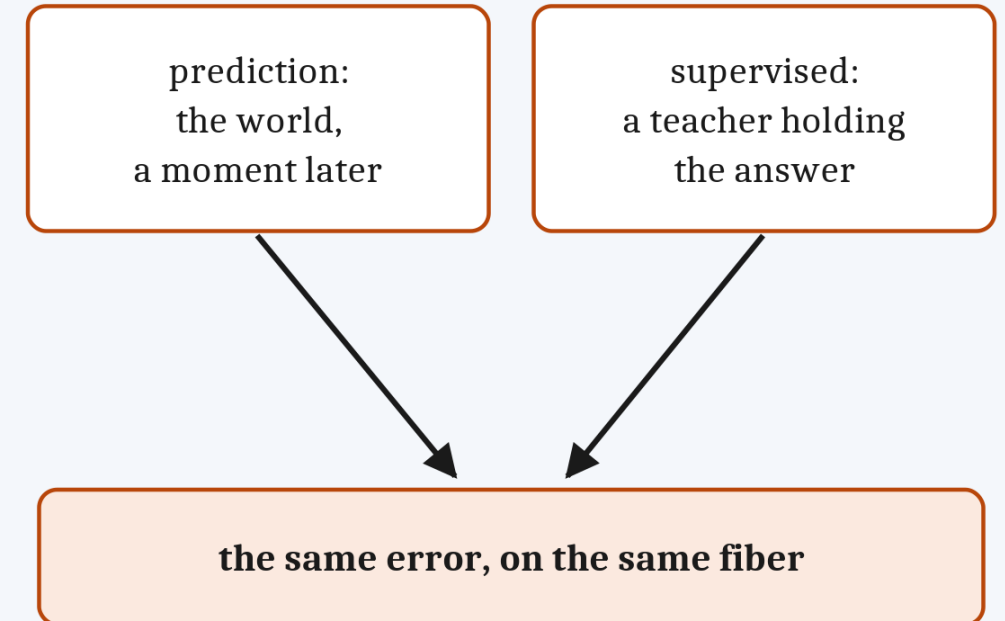
Prediction and *supervised* learning both need an error addressed to one output cell. The cerebellum's climbing fibers deliver exactly that, and cannot say who set the target.

A cerebellar output cell

about 200,000 ordinary inputs



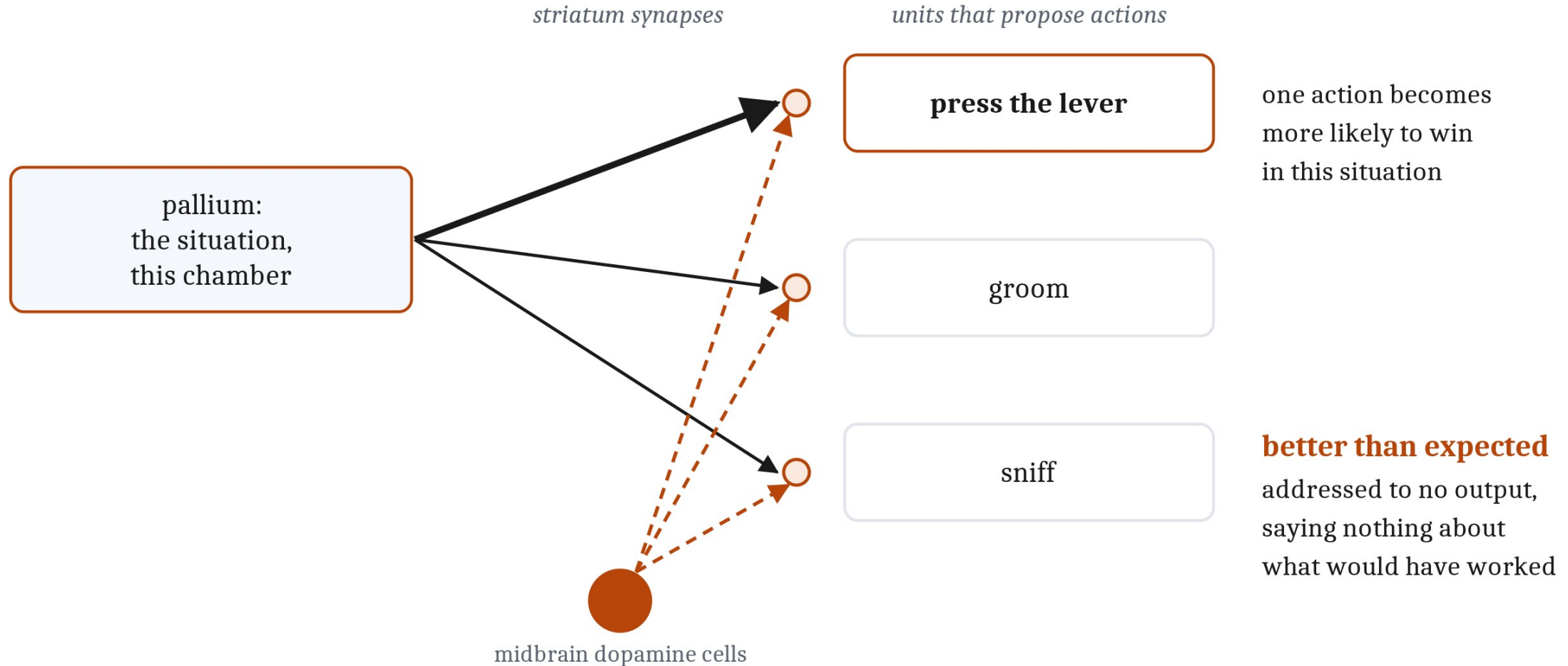
Who set the target?



A synapse cannot tell who supplied the target, so the two families share one structure.

Learning from a Verdict

Reinforcement learning needs only a report of how things went. Dopamine delivers it where the pallium meets the **striatum**, and there it is enough.



Four Families, Three Structures

Two families make the same demand of a synapse. The three mappings are **not equally well supported**, and not exclusive.

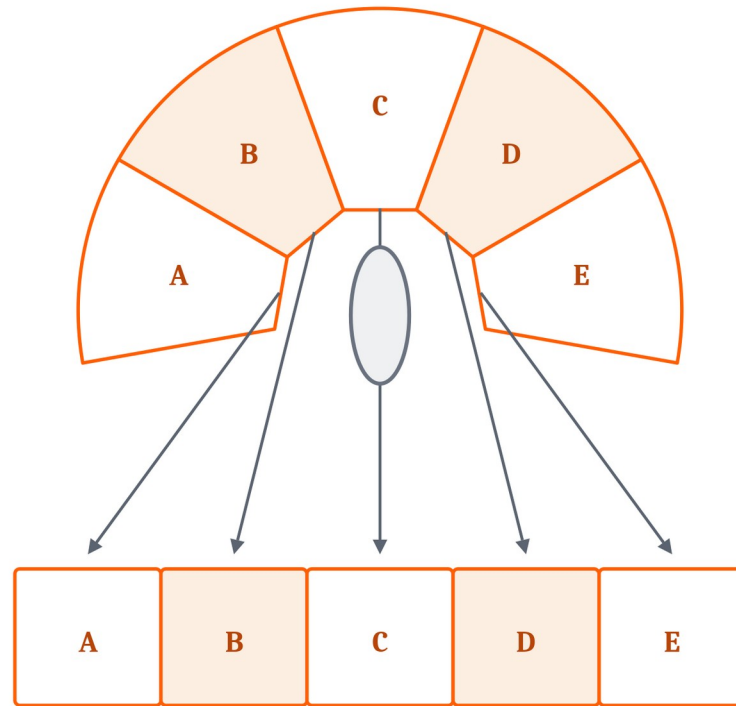
Kind of learning	What has to reach the synapse	Where it comes from	Where it arrives
Unsupervised (Hebbian)	The activity of the two cells the connection joins	Nowhere else; both are already at the connection	Synapses everywhere; Doya's candidate is the pallium
Prediction (self-supervised)	An error, computed separately for each output	The world, a moment later	The cerebellum's climbing fibers, one to each output cell
Supervised	An error, computed separately for each output	A teacher that already holds the answer	The same climbing fibers; the wiring does not tell the two apart
Reinforcement	One number reporting how things went	The body, or the outcome itself	Dopamine fibers from the midbrain, where the pallium meets the striatum

A Few Wiring Plans: Maps and Loops

A few large-scale arrangements, built from Module 4's motifs, recur across the brain. No behavior uses one structure alone.

An ordered copy of the space around the animal

the space around the animal, seen from above

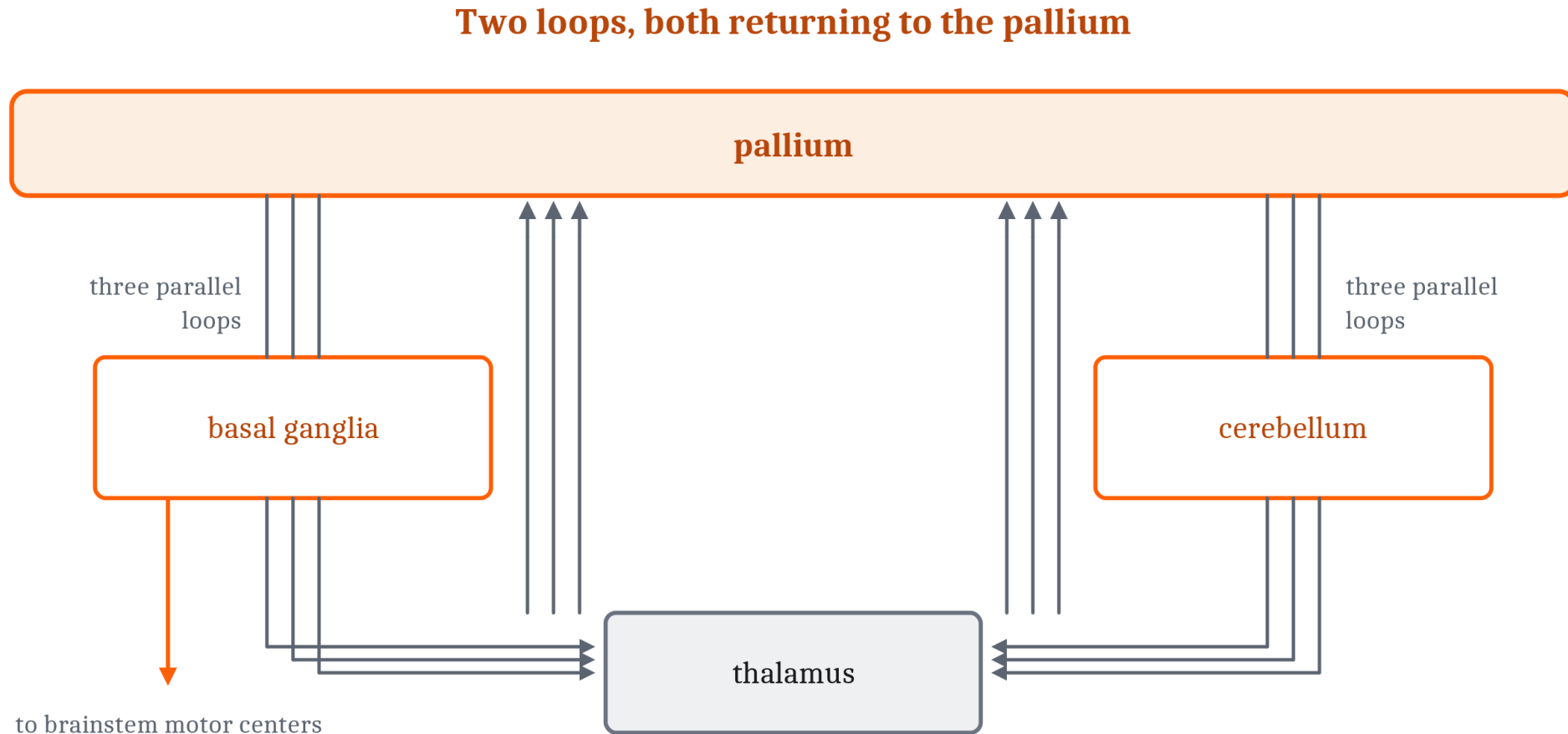


the surface of the tectum

neighboring directions, neighboring patches of tissue — which is what lets the eyes be aimed from it

A Few Wiring Plans: Maps and Loops

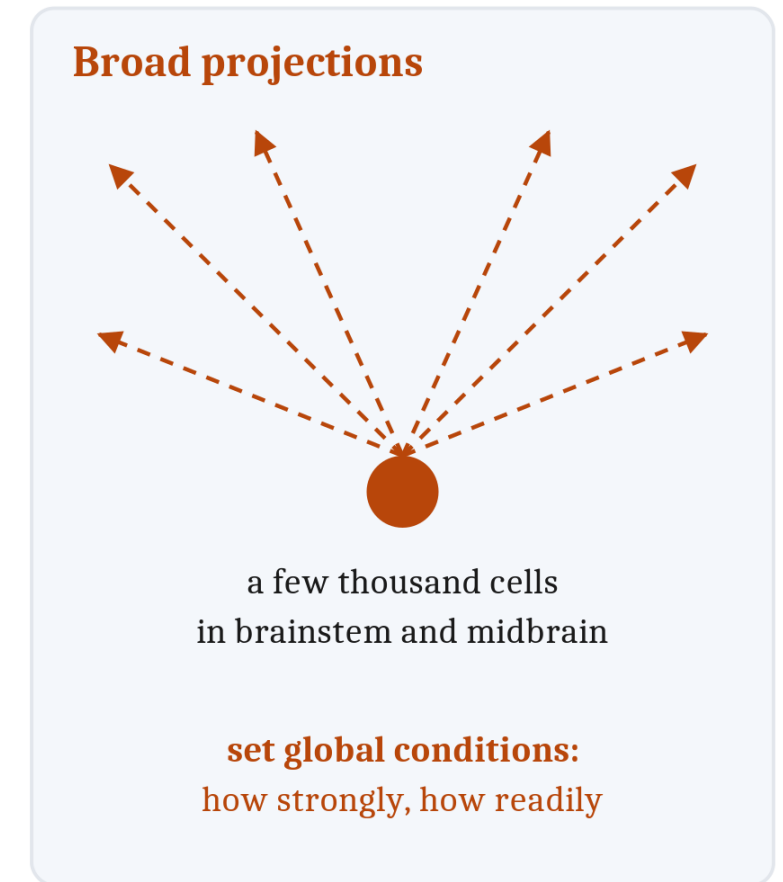
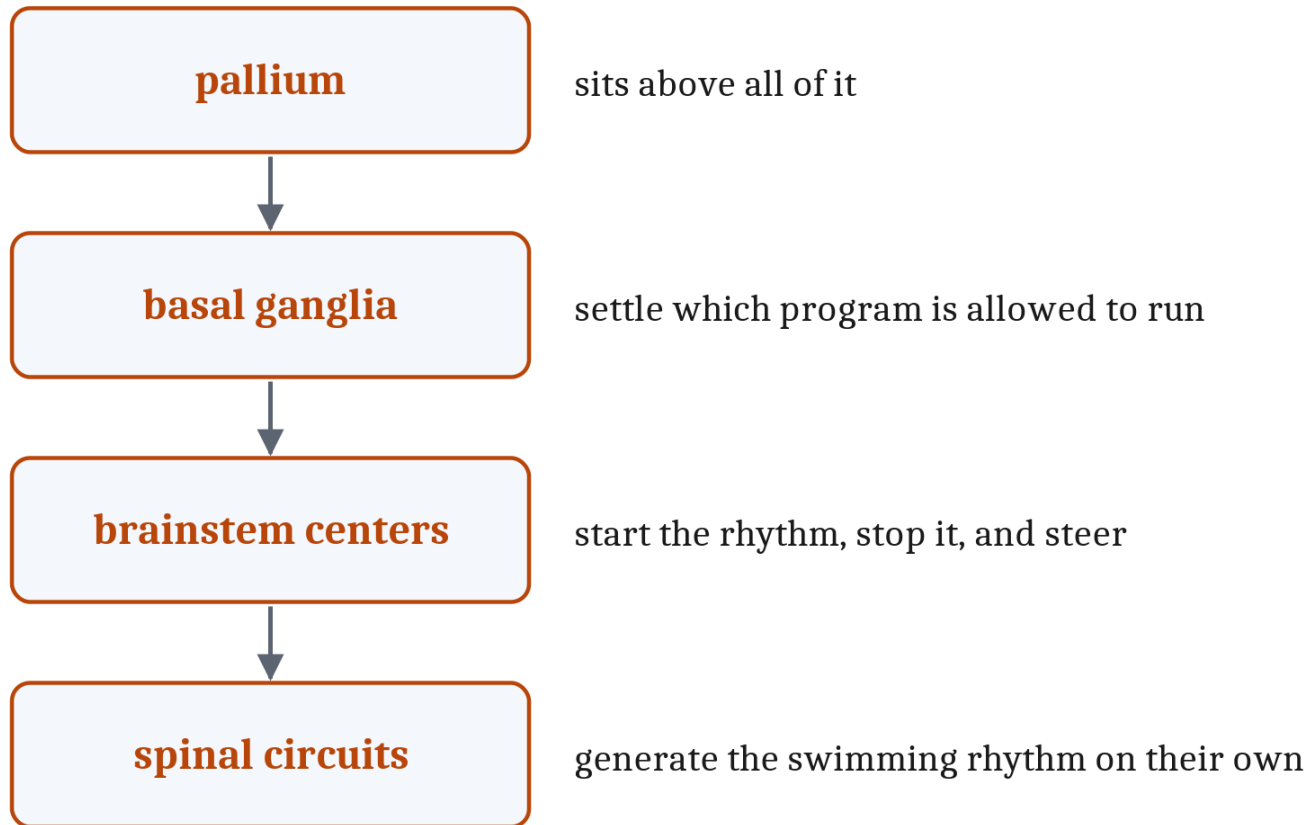
A few large-scale arrangements, built from Module 4's motifs, recur across the brain. No behavior uses one structure alone.



each set of loops leaves and returns to a different stretch of the pallium

Broad Projections, and a Hierarchy of Control

A few thousand cells set conditions for the whole brain. Lower levels run whole behaviors; upper levels choose among them.

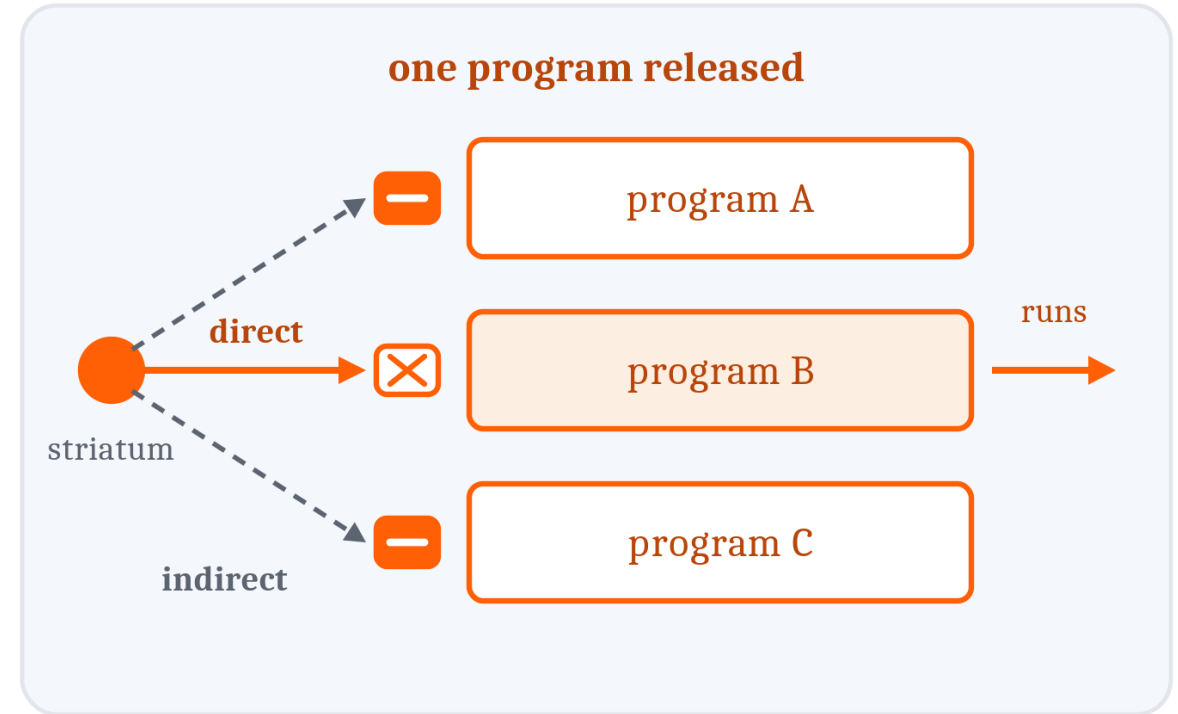


lower levels run whole behaviors; upper levels choose among them and adjust them

Choosing Is Releasing, Not Switching On

Action selection by disinhibition: every program is held back by default. The **direct pathway** releases one; the **indirect pathway** presses the rest down.

Selection by releasing one hold



 hold engaged

 hold released

More Copies of the Same Unit

Modularity: units that do their own job, connect through limited links, and can be copied or altered without redesigning the rest. A claim about wiring, not about faculties of mind.

Lamprey

locomotion
direct + indirect

eye movements
direct + indirect

posture
direct + indirect

chewing
direct + indirect

one selection unit
per motor program

Mammal

locomotion
direct + indirect

eye movements
direct + indirect

posture
direct + indirect

chewing
direct + indirect

another program
direct + indirect

another program
direct + indirect

another program
direct + indirect

not movement
direct + indirect

not movement
direct + indirect

more programs, more copies, some for the pallium's
non-movement work

The design was not improved. It was copied.

Modularity here is a claim about wiring, not about separate faculties of mind.

A Skill That Stopped Needing Attention

In your group, four minutes.

Pick a skill someone in the group practiced until it stopped needing attention: driving, typing, a sport, an instrument.

At each stage of learning it, which family of learning was doing the work, and **what supplied the signal**: nothing, the world, a teacher, or a verdict?

Image Sources (1 of 2)

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